p-adic Analysis 9

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Problem 1

We can define

$$a_n = (n-1)(n-5)b_n$$

where

$$b_n = \begin{cases} 1, & n \equiv 0, 1 \pmod 4, \\ 0, & n \equiv 2, 3 \pmod 4. \end{cases}$$

This sequence satisfies

$$a_{n+12} = 3a_{n+8} - 3a_{n+4} + a_n$$

and the first 12 terms are

$$5, 0, 0, 0, -3, 0, 0, 0, 21, 32, 0, 0.$$

Problem 2

Part (a)

If $y = 0$ then the only solution is $(0, 0)$, so we can set $t = \frac{x}{y} \in \mathbb{Q}$. Dividing the equation by y^2 gives us

$$t^2 - 6t - 6 = 0.$$

Completing the square gives us

$$(t-3)^2 = 15,$$

but $\sqrt{15} \notin \mathbb{Q}$, so f has no nontrivial roots over $\mathbb{Q}$.

Part (b)

By the same argument as before, for f to have a nontrivial root over $\mathbb{Q}_p$, 15 must be a square in $\mathbb{Q}_p$. For $p = 3$ or $p = 5$, $v_3(15) = 1$ and $v_5(15) = 1$. Since it does not have even valuation, 15 is not a square in $\mathbb{Q}_3$ or $\mathbb{Q}_5$.

Problem 4

First, we can rule out odd n. Taking the recurrence mod 3 gives

$$a_n \equiv -5a_{n-2} \equiv a_{n-2} \pmod 3.$$

Since $a_0 \equiv 1$ and $a_1 \equiv 2$, we get

$$a_{2r} \equiv 1 \pmod 3, \quad a_{2r+1} \equiv 2 \pmod 3.$$

So if $a_n = 1$, then n must be even, so we can write $n = 2r$. Now consider the function $f(r) = a_{2r}$.

We can let

$$M = \begin{pmatrix} 0 & 1 \\ -5 & 3 \end{pmatrix}, \quad \begin{pmatrix} a_n \\ a_{n+1} \end{pmatrix} = M^n \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

$$M \equiv \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \pmod 3.$$

$$M^2 = \begin{pmatrix} -5 & 3 \\ -15 & 4 \end{pmatrix} = I + 3B, \quad B = \begin{pmatrix} -2 & 1 \\ -5 & 1 \end{pmatrix}.$$

For $z \in \mathbb{Z}_3$, define

$$F(z) = \text{first coordinate of } (I + 3B)^2 \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Using the binomial expansion,

$$(I + 3B)^2 = \sum_{d=0}^{\infty} \binom{z}{d} 3^d B^d,$$

which converges on $\mathbb{Z}_3$. For $r \geq 0$, this gives us

$$F(r) = a_{2r}.$$

So we need to solve $F(z) - 1 = 0$. Expanding this, we get

$$F(z) - 1 = -27\binom{z}{2} + 81\binom{z}{3} + 486\binom{z}{4} + \cdots.$$

Define $G(z) = \frac{1}{27}(F(z) - 1)$, so that

$$G(z) = -\binom{z}{2} + 3\binom{z}{3} + 18\binom{z}{4} + \cdots,$$

$$G(z) = \frac{z(z-1)(z-3)}{2} + 3R(z),$$

for some $R(z)$. Since the largest 3-adic absolute value happens no later than degree 3, G has at most 3 zeros in $\mathbb{Z}_3$. Trying values, we get that there are zeros at $0, 1, 3$, so these are the only zeros. So $r = 0, 1, 3$ and therefore $n = 2r = \{0, 2, 6\}$.

Problem 9

The proof fails for $p = 2$ because the derivative is $2a_1x_1'$ which is actually $\equiv 0 \pmod 2$, so Hensel's Lemma doesn't apply.

Problem 11

For a fixed prime p, we want neither n nor $n+2$ to be divisible by p. For this to happen, we need $n \not\equiv 0, -2 \pmod p$. In the case of $p \neq 2$, we just need n to be even, so it contributes a probability of $1/2$. For $p > 2$, the probability that n and $n+2$ are not divisible by p is naively $1 - \frac{2}{p}$. Checking through all primes $\leq \sqrt{x}$ gives us a final result of

$$x \cdot \frac{1}{2} \cdot \prod_{2<p\leq\sqrt{x}} \left(1 - \frac{2}{p}\right).$$